Disturbances of vitamin uptake. Pathophysiology of lipid and purine metabolism. Gout.

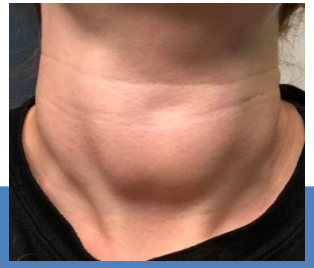
Content

- Nutrients: macronutrients, micronutrients (essential trace elements, minerals and vitamins)
- Lipoprotein metabolism disorders: hyperlipoproteinemia
- Deficiencies in amino acid metabolism: nitrogen excretion disorders, phenylketonuria, homocystinuria
- Disorders of the metabolism of purines, pyrimidines and sugars - hyperuricemia (gout), glycogenoses, galactosemia, fructosemia

Nutrients

1. Macronutrient (fat, protein, carbohydrate)
2. Micronutrients
   - Essential trace elements: zinc (Zn), copper (Cu), selenium (Se), chromium (Cr), cobalt (Co), iodine (I), manganese (Mn), and molybdenum (Mo).
   - Minerals: fluor (F), magnesium (Mg), iron (Fe), potassium (K), chlorine (CL), sodium (Na), calcium (Ca), phosphorus (P)
   - Vitamins

Essential trace elements

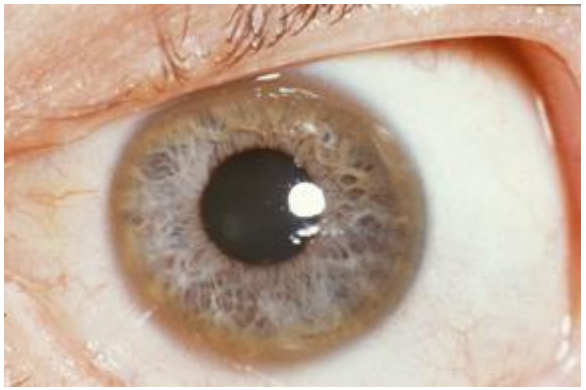


Essential trace elements (a metal cofactor in enzymes)		
Iodine	Required for the synthesis of thyroid hormones	deficiency: goiter, cretinism delayed normal growth and development (in childhood) Excess state: hyperthyroidism
Cobalt	Required for the synthesis of vitamin B12	Deficiency: Pernicious anemia Excess state: cobalt poisoning
Zinc	Zn-bound liver alcohol dehydrogenase, carbonic anhydrase, SOD1 in cytoplasm and extracellular SOD3 (Both SODs have both Zn and Cu) and storage, secretion, and action of insulin	Hypothesis: reduced reproductive function
Copper	Superoxide dismutase (SOD1) in cytoplasm and cytochrome C oxidase (or complex IV contains both Cu and hemes with Fe)	↓ inactivation of ROS , Wilson disease
Manganese	Superoxide dismutase (SOD2) in mitochondria	↓ inactivation of ROS
Molybdenum	The active site of xanthine oxidize with Mo. The enzyme can catalize xanthine to uric acid.	Hypothesis: hyperuricemia in ↑ Mo, gout
Selenium	Se-containing glutathione peroxidase which can protect the organism from oxidative damage	Hypothesis: ↓Se in hypothyroidism

Wilson disease

- An inherited disorder (autosomal recessive pattern) in which excessive amounts of copper accumulate in the body: the liver (1st) than brain and eyes.
  - diminished biliary excretion of copper
  - failure to insert copper into ceruloplasmin (a glycoprotein that transports copper in the blood)
- Excesses of copper generate free radicals that disrupt cellular organelles
  - clumsiness, tremors, difficulty walking, speech problems, impaired thinking ability, depression, anxiety, and mood swings.
  - mild liver disease: yellowing of the skin or of the eyes (jaundice), cirrhosis
  - green-to-brownish ring, called the Kayser-Fleischer ring

dark rings that appear to encircle the iris of the eye



Minerals

Potassium	A systemic electrolyte and is essential in coregulating ATP with sodium	hypokalemia	hyperkalemia
Chlorine	Needed for production of hydrochloric acid in the stomach and in cellular pump functions	hypochloremia	hyperchloremia
Sodium	A systemic electrolyte and is essential in coregulating ATP with potassium	hyponatremia	hypernatremia
Calcium	Needed for muscle, heart and digestive system health, builds bone, supports synthesis and function of blood cells	hypocalcaemia	hypercalcaemia
Phosphorus	A component of bones (see apatite), cells, in energy processing, in DNA and ATP (as phosphate) and many other functions	hypophosphatemia	hyperphosphatemia
Magnesium	Required for processing ATP and for bones	magnesium deficiency	hypermagnesemia
Iron	Required for many proteins and enzymes, notably hemoglobin to prevent anemia	iron deficiency	iron overload disorder

Hemosiderosis

- An excessive iron is stored as hemosiderin in the cells of many organs and tissues
- Causes:
  - repeated blood transfusions
  - hemolytic anemia
  - excessive alcohol ingestion
  - **Hereditary hemochromatosis:** a genetic disorder of iron metabolism and the most severe example of iron overload caused by excessive intestinal absorption of elemental iron.
    - Manifestation:
      - liver (cirrhosis)
      - pancreatic cell (bronze diabetes)
      - bronze pigmentation of the skin

Vitamins

- Fat-soluble (DEKAF): They are stored (mainly D and A vitamins) and their shortage lead to late signs. Their excess leads to toxic hypervitaminoses (e.g. vitamin A: hepatomegaly, vomiting, bone and joint pain)

- Water-soluble vitamins (8 B vitamins and C vitamin): Except for B12 vitamin, they are not stored and their shortage lead to yearly signs.

Thiamine (vitamin B1) deficiency

- Severe and chronic deficiency lead to beriberi
- Causes
  - alcoholics (vitamin B1 is stored in the liver)
  - polished rice

Two forms of beriberi

- Wet beriberi affects the cardiovascular system resulting in cardiomyopathy (dilated cardiomegaly, biventricular heart failure): hyperdynamic circulation (systemic vasodilation that is compensated by high cardiac output and fast heart rate, shortness of breath), and leg swelling.

- Dry beriberi affects the nervous system by demyelination resulting in peripheral neuropathy and than muscle wasting
  - Tingling or numbness of the hands and feet and partial paralysis
  - Symmetrical and neuropathy: difficulty rising from squatting position
  - Confusion and pain
  - Impaired sensory perception and motor weakness, poor reflexes

Dry beriberi: Wernicke encephalopathy

Wernicke encephalopathy (CNS) is characterized by the triad

- ophthalmoplegia: refers to weakness (-paresis) or paralysis (-plegia) of one or more extraocular muscles which are responsible for eye movements and later expanded to other eye movement disorders (nystagmus)
- ataxia: lack of voluntary coordination of muscle movements
- confusion

Case (video)

- Description
- A 34-year-old man was presented to our emergency department in alcohol withdrawal.
- Despite initial treatment with high-dose intravenous thiamine therapy he went on to develop nystagmus , ataxia, pass pointing, intention tremor and worsening confusion. He was diagnosed with Wernickes encephalopathy, an acute neuropsychiatric syndrome resulting from thiamine deficiency
- His serum magnesium levels were found to be low at 0.41 mmol/L (normal range 0.66–1.02 mmol/L). He was started on high-dose intravenous magnesium in addition to thiamine replacement and his neurological symptoms resolved once his serum magnesium levels had normalised
- https://casereports.bmj.com/content/2016/bcr-2016-218046

Dry beriberi: Korsakoff syndrome

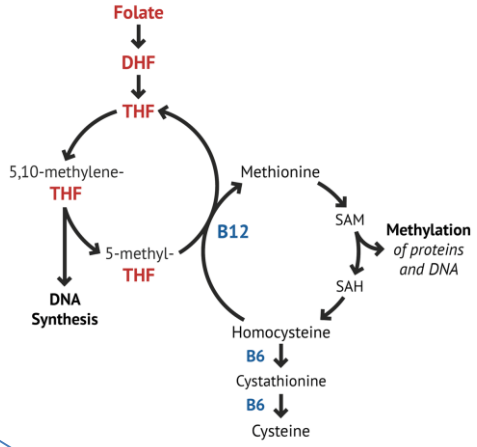
- is often accompanied by Wernicke encephalopathy
- an amnestic disorder: deficit in memory caused by brain damage.
  - confabulation, that is, invented or fabricated memories which are then taken by the patient as true due to gaps in memory, with such gaps sometimes associated with blackouts
- hallucination
- minimal content in conversation
- apathy

Vitamin B_{12} (cobalamin) deficiency

- It is stored in the liver (storage for several years, not very common in young adults)
  - older people are more prone to this condition leading to peripheral neuropathy, neurodegenerative disorder or psychiatric problems).
  - in fetal and early life cause poor brain development and cognitive functions
- Causes:
  - malabsorption due to intrinsic facto (IF) deficiency
    - atrophic gastritis (in the elderly)
    - surgical resection
    - intrinsic facto antibodies (pernicious anemia) often combination with Graves' disease (hyperthyroidism) and other autoimmune disorders
  - Inadequate dietary intake in vegans
  - Genetic diseases: common mutations in methylenetetrahydrofolate reductase (MTHFR) gene

Vitamin B_{12} (cobalamin): biochemistry

Hyperhomocysteinemia is regarded as a marker of cardiovascular disease (blood clots, heart attacks, and strokes), likely working through atherogenesis



- Neurological manifestation due to low methionine for myelin sheaths

- conversion of homocysteine to methionine
- conversion of methylmalonal-CaA to succinyl-CoA
- conversion of 5-methyltetrahydrofolate to tetrahydrofolate

- A lack for neurotrans mitter synthesis

Limited DNA synthesis leading to megaloblastic anemia and others resulting from the cells with high turnover (GIT)

Niacin (vitamin B3) deficiency

- Pellagra (also known as “the three Ds”) is a disease marked by dementia, diarrhea and dermatitis.
- If left untreated, pellagra can be fatal.





seborrheic dermatitis: is a common skin condition that causes scaly patches, red skin and stubborn dandruff.

Vitamin B6 deficiency (pyridoxine)

- Signs:
  - peripheral neuropathy and a pellagra-like syndrome with seborrheic dermatitis
  - glossitis and cheilosis
  - depression, confusion, seizures
- Causes:
  - Alcoholism
  - Hemodialysis
  - pyridoxine-inactivating drugs (e.g. penicillamine)



Cracked corners of the mouth in cheilosis and glossitis (inflammation with loss of the lingual papillae)

Vitamin C deficiency

- an antioxidant
- facilitates the absorption of iron
- collagen formation and scurvy in advanced deficiency (bleeding sores, tooth loss, anemia, and a reduced rate of healing for injuries).

Vitamin D deficiency

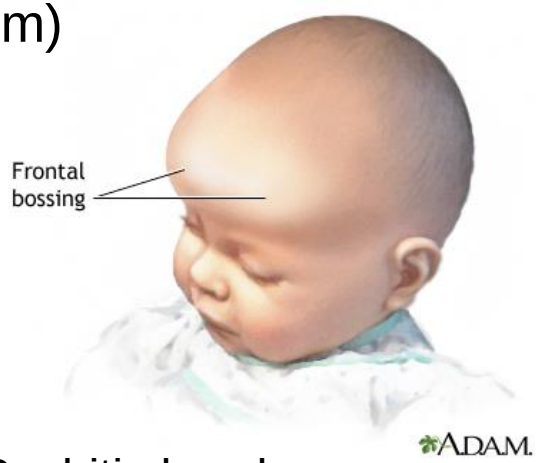
- caused by
  - inadequate nutritional intake of vitamin D
  - disorders limiting vitamin D absorption
    - pancreatic insufficiency,
    - celiac disease
    - Cholestasis(failure of bile flow)
  - disorders impairing vitamin D conversion into active metabolites—including certain liver, kidney
  - inadequate sunlight exposure (UV)

Vitamin D deficiency



- In childhood: rachitis
  - bone deformities
    - craniotabes (softening or thinning of the skull),
    - outward bowed legs and hips
    - rosary or beading of the ribs
    - weak or soft bones leads to often fractures
- In adult: osteomalacia
  - weak and soft bones
  - muscle weakness
  - joint and bone pain (secondary hyperparathyroidism)





Rachitic beads

Vitamin D deficiency

- Biochemical findings:
  - low serum and urinary calcium
  - low serum phosphate, except chronic kidney disease (decreased glomerular filtration leads to P accumulation that secondary decrease Ca^{2+})
  - elevated serum alkaline phosphatase due to an increase in compensatory osteoblast activity (Note: High levels of ALP is also indicate liver disease)
  - elevated parathyroid hormone (due to low calcium)

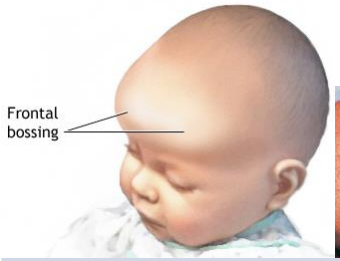
Osteocalcin is vitamin K-dependent and a marker of bone formation (note: it correlates with increases in bone mineral density) that also increases insulin sensitivity in many tissues

Table 1
Biochemical markers of bone turnover.

Bone formation	Bone resorption
Bone alkaline phosphatase (BAP)	Pyridinium crosslinks
Osteocalcin (OC)	Pyridinoline (PYD)
Type I collagen extension propeptides (PICP, PINP)	Deoxypyridinoline (DPD)
	Crosslinking telopeptides of type I collagen
	C-terminal (CTX) – N-terminal (NTX)
	Tartrate resistant acid phosphatase (TRAP)
	Galactosyl-hydroxylysine
	Hydroxyproline

Vitamin D

hormone	Response to	serum	bone	GIT	kidney
Parathyroid hormone parathyroid gland	↓ Ca^{2+}	hypercalcemia	↓	via calcitriol	↑ reabsorption of Ca^{2+} but excretion of P and HCO_3^-
Calcitriol (from cholesterol in the skin, than kidney and liver)	parathyroid hormone	Depends on concentration of calcitriol		↑absorption of Ca^{2+} and P in same ratio (independent on parathyroid hormone)	↑ reabsorption of Ca^{2+} and P
Calcitonin thyroid gland	Acute (not chronic) hypercalcemia	hypocalcemia	short-term bone calciphication	↓absorption of Ca^{2+} and P in same ratio	↑excretion of both Ca^{2+} and P



Rickets: Rachitic beads



- The overall effect of PTH secretion is to increase serum calcium concentration and decrease serum phosphate level.

Vitamin E deficiency

- Causes: Fat malabsorption
- Vitamin E is an antioxidant.
- Sign:
  - neurological and neuromuscular problems: poor conduction of electrical impulses along nerves due to changes in nerve membrane structure and function (the damaging effects of free radicals).
  - hemolytic anemia – due to oxidative damage to red blood cells.

Vitamin K deficiency

- Causes:
  - hepatic failure
  - vitamin K_1-antagonist such as warfarin
  - disruptions of colonic bacteria (antibiotics and newborns)

- Colonic bacteria synthesize a significant portion of the Vitamin K required for human needs and only small amount in food is needed. Vitamin K is changed to its active form in the live.

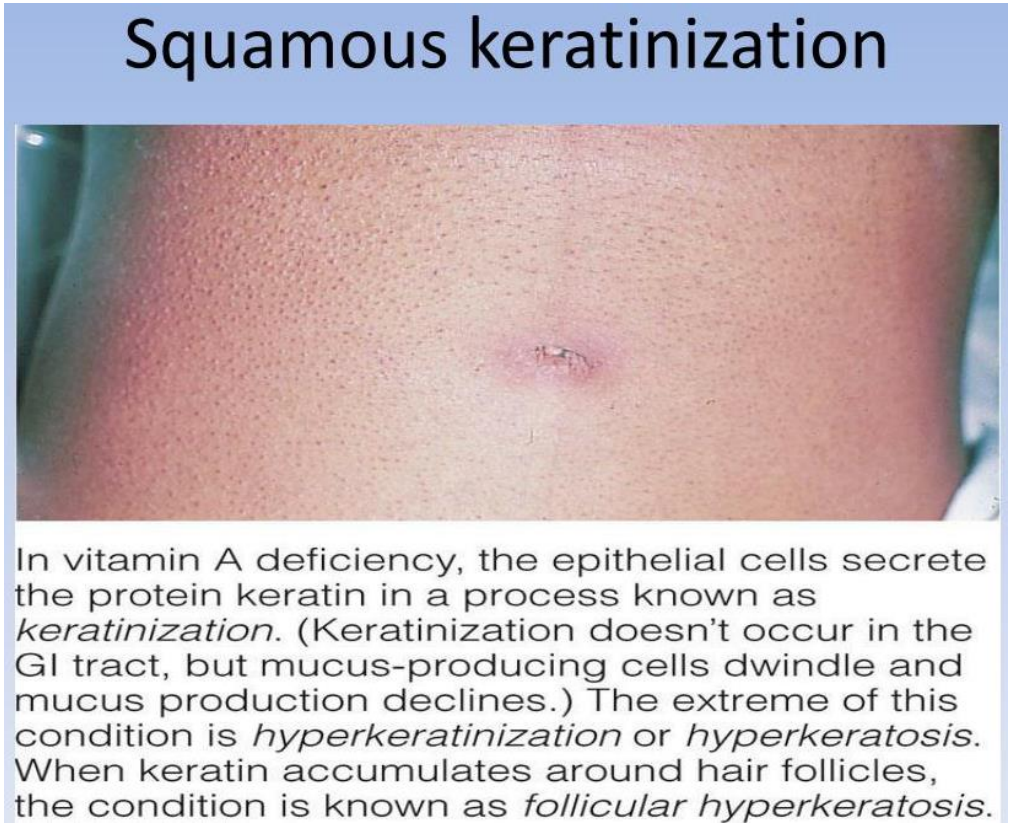
- Clotting factors II, VII, IX, X, and anti-coagulant proteins (protein C and protein S) and osteocalcin.

- Signs: bruising, petechiae, hematomas and massive or chronic uncontrolled bleeding (nose ,GIT, brain)

- Prothrombin time (PT or quick test) is elevated, but the partial thromboplastin time (PTT) is normal or only mildly prolonged because factor VII has the shortest half-life.

Vitamin A deficiency

- Dark blindness
- Squamous epithelium due to keratinization (abnormal quantity of keratin):
  - Skin, upper respiratory passage and urinary tract



Disorders of the metabolism of purines, pyrimidines

- Hyperuricemia and gout: DNA and uric acid
- Porphyria: heme synthesis

Gout: deposits of urate crystals

- Uric acid is a breakdown product of purine nucleotides (DNA)

- Cause (hyperuricemia):
  - Overproduction of uric acid:

    Accelerated rate of purine synthesis accompanied by an overproduction of uric acid or some patients break down purine nucleotides
  - Underexcretion of uric acid:

- Pathophysiology: In gout, undigested uric acid accumulates within lysosomes, damaging the lysosomal membrane. Subsequent enzyme leakage results in cell death and tissue injury (the inflammation).

Gouty tophus

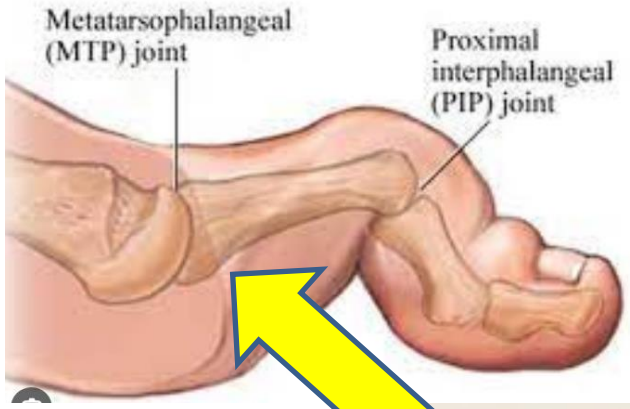


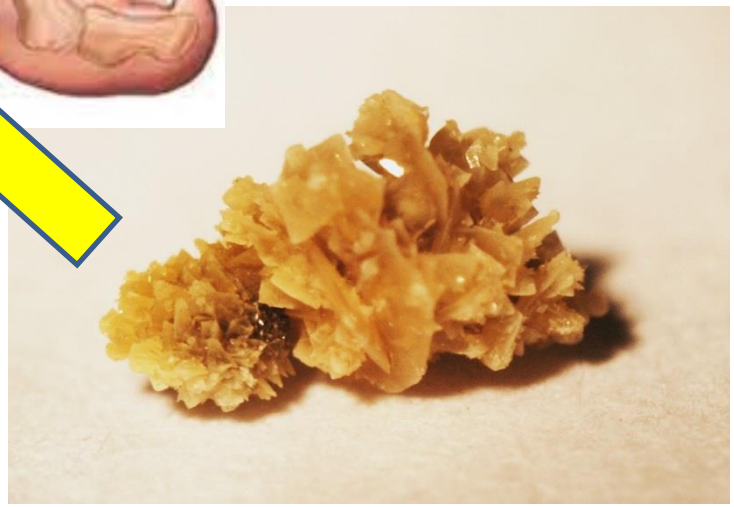
Gout: typical manifestation

- Decreasing temperatures cause both urate and uric acid solubility to fall.
- Lead poisoning can interfere with excretion of urate and predispose to gout.



Tophaceous deposits

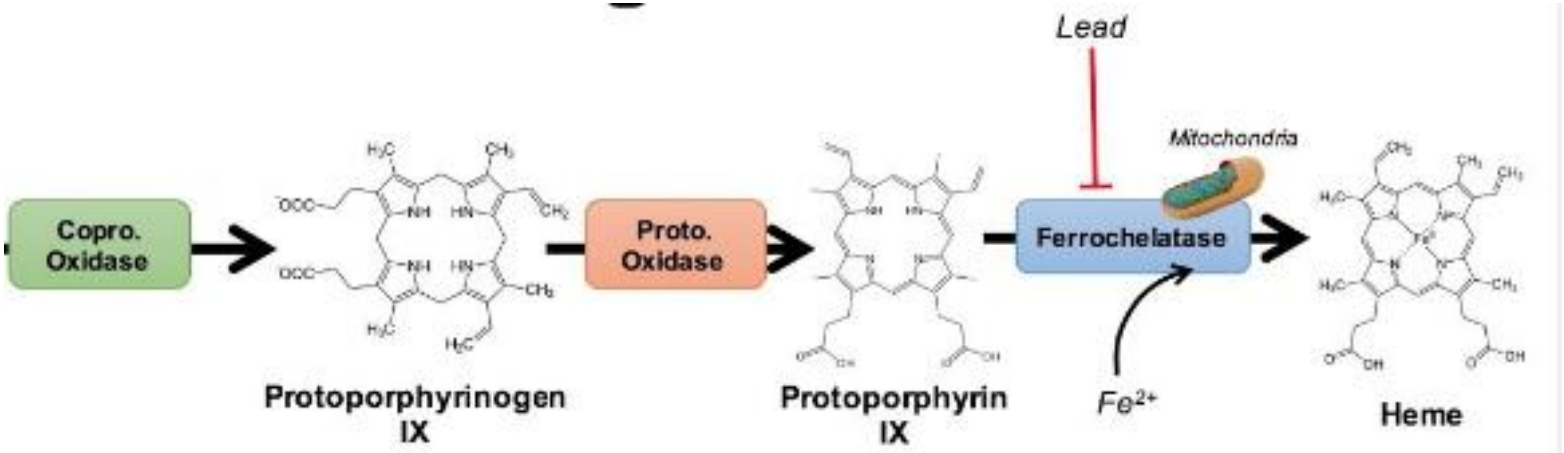






Porphyrias

- Porphyrias are metabolic disorders of the heme biosynthesis
- Porphyrins build up in the body and affect the liver or erythropoiesis
- Two types
  - hepatic porphyria: accumulation of porphirins in the liver
    - acute intermittent porphyria
    - porphyria cutanea tarda
  - erythropoietic porphyria: accumulation of porphirins in the bone marrow and erythrocytes
    - congenital erythropoietic porphyria = Gunther disease extremely rare: severe photosensitivity with erythema, swelling, blistering, hemolytic anemia, splenomegaly



Heme synthesis pathway

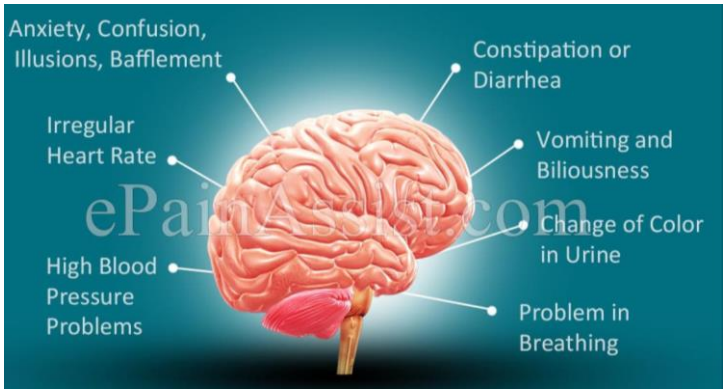
Acute porphyrias:
acute intermittent porphyria (AIP)

- An autosomal dominant porphyria resulting in about 50% normal activity of the affected enzyme.

- Intermittent: The triggers for acute attacks can be medications.

- Its manifestation is an acute attack: The exact mechanism of acute attacks is not clear. One hypothesis is a toxic effects on neurons. The major symptom of an acute attack is abdominal pain, often accompanied by vomiting, hypertension, and tachycardia. Severe attack: motor neuropathy: quadriplegia (paralysis of all four limbs) central nervous system: seizures and coma.
  - 90% of affected individuals never experience an acute attack and are asymptomatic

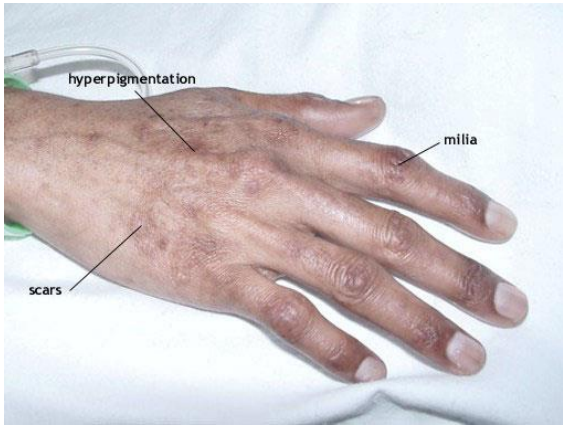
- The absence of photosensitive



Cutaneous/chronic porphyrias: porphyria cutanea tarda (PCT)

- the most common type of porphyria
  - PCT is due to underlying liver disease (excessive alcohol consumption hepatitis B and C) or inherited in an autosomal dominant pattern (only 20%)
  - A primary cause: uroporphyrinogen decarboxylase deficiency (UROD: one of the enzyme responsible for heme production.).

- primary manifestation is with skin disease (cutaneous porphyrias)
  - Porphyrins are photoactive molecules: exposure to light results in promotion of electrons to higher energy levels. When these return to the resting energy level or ground state, energy is released. This causes local skin damage.
  - fluid or blood-filled blisters, ostinflammatory pigmentation and cysts (milia)



Dyslipidemia (or dyslipoproteinemia)

- Abnormal concentrations of serum lipoproteins

- Causes:
  - Primary dyslipoproteinemias result from:
    - genetic defects (familial dyslipoproteinemias)
    - dietary factors
  - Secondary dyslipoproteinemias result from diabetes, hypothyroidism, pancreatitis, renal nephrosis and medications such as certain diuretics, beta-blockers or glucocorticoids.

Familial hypercholesterolemia (FH)

- autosomal dominant disorders with various genetic defects in LDL receptor
- accelerated atherosclerosis, coronary disease and fatal myocardial infarction
- cholesterol deposits in skin and tendons (*xanthomas)*
- plasma cholesterol levels are approximately twice as high as normal

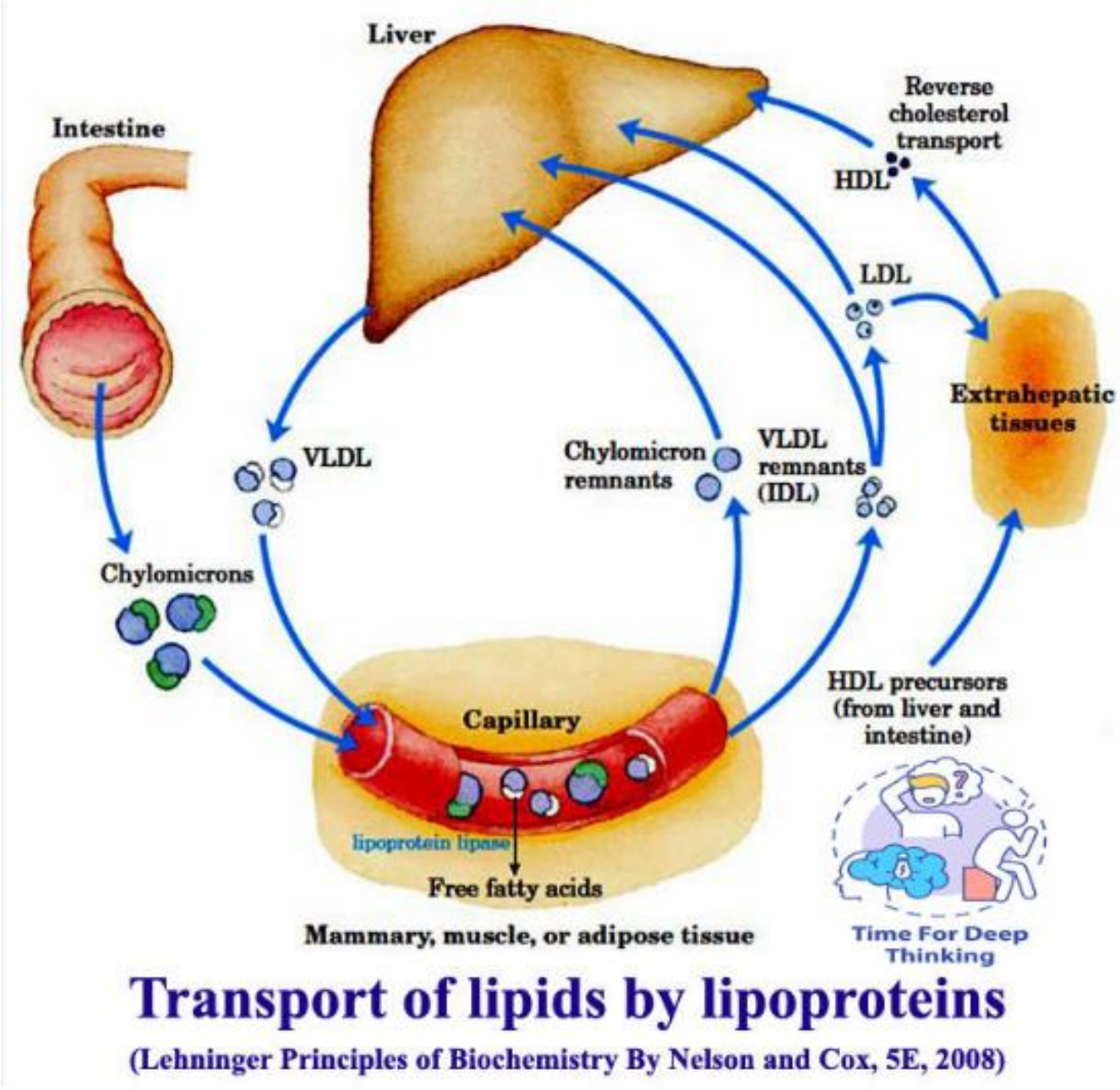
TABLE 32-6 FAMILIAL DYSLIPOPROTEINEMIAS

NAME	LABORATORY FINDINGS	CLINICAL FEATURES	THERAPY
Type I: exogenous hyperlipidemia; fat-induced hypertriglyceridemia	Cholesterol normal Triglycerides increased three times Chylomicrons increased	Abdominal pain Hepatosplenomegaly Skin and retinal lipid deposits Usual onset: childhood	Low-fat diet
Type IIa: hypercholesterolemia	Triglycerides normal LDL increased Cholesterol increased	Premature vascular disease Xanthomas of tendons and bony prominences Common Onset: all ages	Low-saturated-fat and low-cholesterol diet Cholestyramine[a] Colestipol[b] Lovastatin[c] Nicotinic acid[d] Neomycin[e] Intestinal bypass
Type IIb: combined hyperlipidemia; carbohydrate-induced hypertriglyceridemia	LDL, VLDL increased Cholesterol increased Triglycerides increased	Same as IIa	Same as IIa; *plus* carbohydrate restriction Clofibrate[f] Gemfibrozil[g] Lovastatin
Type III: dysbetalipoproteinemia	IDL or chylomicron remnants increased Cholesterol increased Triglycerides increased	Premature vascular disease Xanthomas of tendons and bony prominences Uncommon Onset: adulthood	Weight control Low-carbohydrate, low-saturated-fat, and low-cholesterol diet Alcohol restriction Clofibrate Gemfibrozil Lovastatin Nicotinic acid Estrogens[h] Intestinal bypass
Type IV: endogenous hyperlipidemia; carbohydrate-induced hypertriglyceridemia	Glucose intolerance Hyperuricemia Cholesterol normal or increased VLDL increased Triglycerides increased	Premature vascular disease Skin lipid deposits Obesity Hepatomegaly Common onset: adulthood	Weight control Low-carbohydrate diet Alcohol restriction Clofibrate Nicotinic acid Intestinal bypass
Type V: mixed hyperlipidemia; carbohydrate and fat-induced hypertriglyceridemia	Glucose intolerance Hyperuricemia Chylomicrons increased VLDL increased LDL increased Cholesterol increased Triglycerides increased three times	Abdominal pain Hepatosplenomegaly Skin lipid deposits Retinal lipid deposits Onset: childhood	Weight control Low-carbohydrate and low-fat diet Clofibrate Lovastatin Nicotinic acid Progesterone[i] Intestinal bypass

Dyslipidemia (or dyslipoproteinemia): classification

- Hyperlipidemia is classified according to which types of lipids are elevated:
  - hypercholesterolemia: The UK National Health Service recommends upper limits of total cholesterol of 5 mmol/L, and low-density lipoprotein cholesterol (LDL) of 3 mmol/L.
  - hypertriglyceridemia
  - combined hyperlipidemia.

Lipoprotein metabolism disorders



Lipoprotein

- A **lipoprotein** is an assembly whose primary purpose is to transport hydrophobic lipid molecules in water (blood).

- A lipoprotein consists of
  - lipid center (triglycerides and cholesteryl esters = a dietary lipid, is an ester of cholesterol)
  - outer shell, with the hydrophilic portions oriented outward toward the plasma and lipophilic portions oriented inward toward the lipid center. (phospholipids, cholesterol and embedded apolipoproteins that determines its fate).

Exogenous pathway of Lipoproteins

- Chylomicrons transport exogenous (dietary) products from the intestines to other locations in the body.

- Nascent chylomicron*:* The enterocytes use fatty acids and monoacylglycerides from the chime and transform again into triglycerides and add apolipoprotein B-48

- Mature chylomicrons: Nascent chylomicrons interact with HDL particles, resulting in HDL donation of apolipoprotein C-II and apolipoprotein E to the nascent chylomicron.
  - Via apolipoprotein C-II, mature chylomicrons activate lipoprotein lipase (LPL), an enzyme on endothelial cells lining the blood vessels in the adipose (mainly) and muscle (less) tissues. LPL catalyzes the hydrolysis of triglycerides that releases glycerol and fatty acids.
  - They are absorbed in peripheral tissues (adipose and muscle).

- Chylomicron remnants: When a large portion of the triglyceride core has been hydrolyzed, chylomicron remnants are formed and are taken up by the liver (apolipoprotein E).

Endogenous pathway of Lipoproteins

- **Very-low-density lipoprotein** (**VLDL**): is a type of lipoprotein made by the liver (e.g. after the excessive incomes of sugar or ethanol). VLDL transports endogenous triglycerides and circulates in blood. Lipoprotein lipase (LPL) removes the triglycerides from VLDL for storage or energy production.

- Intermediate-density lipoprotein (IDL) or VLDL remnants: As more and more triglycerides are removed from the VLDL, it becomes IDL. Via an interaction between apolipoprotein E and the remnant receptor, IDL are absorbed by the liver or escape to produce LDL.

- IDL remnants called low-density lipoproteins (LDL): If their cholesterol content becomes greater than the content of triglyceride, IDL becomes LDL. The majority of LDL content (cholesterol) is either stored, secreted in bile (free cholesterol) used for cell membrane structure (cholesterol in the cellular memrane), or converted into other products such as steroid hormones or bile acids. Macrophages remover small part of the circulating LDL.

Endogenous pathway II

- HDL is produced by the liver. It is the densest because it contains the highest proportion of protein to lipids.
  - HDL remove the tissue cholesterol from e.g. dead cells and transfer it to VLDL and then LDL .
  - HDL is able to prevent the process of oxidation of LDL and the subsequent inflammatory processes seen after oxidation.
  - HDL has the ability to bind LPS, creating HDL-LPS complexes to neutralize the harmful effects in the body and clear the LPS from the body.

- LDL contains apolipoprotein B (apoB), which allows LDL to bind to different tissues, such as the artery wall
  - if they have been damaged by high blood sugar levels.
  - if oxidised, the LDL can become trapped in the proteoglycans, preventing its removal by HDL cholesterol efflux. Normal functioning.

Defects in Amino Acid Metabolism

- An enzymatic deficiency are associated with increased blood concentrations of the amino acid whose degradation pathway is impaired and with the appearance of the amino acid in the urine.

- Examples:
  - Phenylketonuria (PKU)
  - Homocystinuria (HCU)

Homocystinuria (HCU)

- Genetic defect in enzymes that convert homocysteine to methionine or other of the amino acids (an inherited autosomal recessive trait)

- Pathophysiology: an accumulation of the amino acid homocysteine in the serum

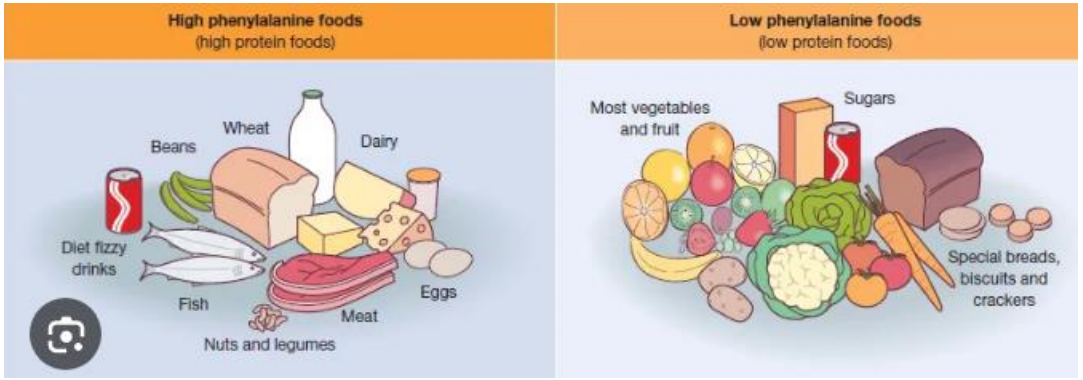
- Signs and symptoms:
  - Intellectual disability
  - Musculoskeletal diseases: osteoporosis, abnormally-shaped chest and bone and joint problems
  - Vascular diseases: Homocysteine binds to the endothelium of the blood vessels and activate signaling pathways leading to the release of pro-inflammatory molecules. This may cause the following :
    - Extensive atheroma formation at a young age which affects many arteries but not the coronary arteries
    - Intravascular thrombosis

- Symptoms of homocystinuria can also be caused by a deficiency of vitamins B6, B12, or folate

Phenylketonuria (PKU):

- Mutations of the phenylalanine hydroxylase (*PAH) gene* results in phenylalanine hydroxylase deficiency and the inability of the body to convert the essential amino acid phenylalanine to tyrosine, leading to the accumulation of phenylalanine in the serum.
  - The accumulation of phenylalanine: brain growth slow, intellectual disability, mental retardation, jerking movements, seizures, skin rashes and attention problems.
  - The lack of tyrosine : blond hair, blue eyes, and fair skin (lack of melanin)







Disorders of the metabolism of sugars: hypoglycemia

- **Hypoglycemia: <3 mmol/L (more dangerous than hyperglycemia,** first manifestation is activation of sympathetic NS and advanced hypoglycemia leads to cerebral edema and coma**)**
- Postprandial hypoglycemia (after food)
  - Alimentary hyperinsulinism: After GIT operation, some people with rapid stomach emptying leading to fast release of insulin
  - **Galactosemia:** a rare genetic metabolic disorder preventing glucose release
  - **Fructosemia:** fructose intolerance, genetic disease an accumulation of fructose-1-phosphate which, over time, results in the death of liver cells, jaundice + hypoglycemia (fructosa-1phosphate inhibits gluconeogenesis)
- Hypoglycemia with low production of glucose
  - Enzymopathy: **glycogenosis**
  - Endocrinopathy: hypopituitarism or hypocorticalism (Addison disease)
  - Sever malnutrition
  - Alcohol and drugs
- Excessive glucose consumption
  - Excessive energetic demand: normoinsulinism
  - Excessive effect of insulin: hyperinsulinism
    - Insulinoma: a rare tumor of endocrine cells in pancreas
    - **Treatment of Type I and II diabetes mellitus**

Disorders of the metabolism of sugars: hyperglycemia

- Hyperglycemia: > 7mmol/L

- Manifestation:
  - Glycosurie if serum glucose >10mmol/L, dehydration
  - Acute: Hyperglycemic and hyperosmolar coma in Type II diabetes mellitus
  - Chronic: AGE advanced glycation endproducts damage vessel wall, hemoglobin glycation, polyol pathway increasing sorbitol in some organs, ROS production
    - The 1st step in the sorbitol pathway is the conversion of glucose to sorbitol. The 2nd step in this pathway is the conversion of sorbitol into fructose via the enzyme sorbitol dehydrogenase. A point of concern is that not all tissues have the enzyme sorbitol dehydrogenase, and these include the retina, kidneys, and Schwann cells. As a result, diabetic patients with ↑ glucose levels accumulate sorbitol to toxic levels within these organs.

- Causes of hyperglycemia:
  - Diabetes mellitus Type I
  - Diabetes mellitus Type II
  - Insulin resistance that accompany many diseases, e.g. infection
  - Increased corticoids: Cushing‘s syndrome or drugs
  - Adenoma of pituitary gland: acromegaly (STH)
  - Glucagonoma: a very rare tumor of the pancreatic alpha cells that results in the overproduction of the hormone, glucagon.

References

Look up the particular theme in the following indexes

- Porth's Pathophysiology: Concepts of Altered Health States (9th Edition) by Sheila Grossman, Carol Mattson Porth. Wolters Kluwer Health | Lippincott Williams & Wilkins, 2014.
- McCance K. L., Huether S. E.: Pathophysiology. 8th Edition. Mosby, 2018.
- Silbernagl S, Lang F. Color Atlas of Pathophysiology, 3rd Ed. Thieme, 2016.